

RAVISHEK KUMAR

Bachelor's in Mechanical Engineering and Master's in Financial Engineering

Financial Engineering student at IIT Kharagpur researching behavioral household finance and quantitative methods, with experience in econometric analysis of large-scale survey data, alpha generation, and time-series forecasting.

ravishekkendra@gmail.com | +91-7000729959 |  LinkedIn |  GitHub

Education

Indian Institute of Technology Kharagpur

Aug 2022 – May 2027

Dual Degree (B.Tech in Mechanical Eng. & M.Tech in Financial Eng.)

CGPA: 8.96/10

Experience

Fidelfolio Investments

May 2026 – Jul 2026

Quantitative Investment Internship

Mumbai, India

- Engineered an **automated pipeline** to generate mutual fund research reports and investment booklets, seamlessly integrating **financial visualizations, tables, and performance summaries**.
- Streamlined **quantitative research workflows** for long-term investment products, significantly reducing manual publication effort and ensuring **structural consistency**.
- Authored the industry paper, “**The Last Mile of Quantitative Research: Automating Charts, Narratives and PDF Publication**,” detailing the **end-to-end automated reporting framework**.

Projects

Behavioral Household Finance in India: Evidence from CMIE Consumer Pyramids Survey

Aug 2026

Independent Research Project

- Engineered a reproducible **Python/DuckDB data pipeline** to merge four **CMIE Consumer Pyramids modules**, constructing a **~480,000-observation household panel** for 2025.
- Evaluated **Laibson's present bias** and the “co-holding” of high-cost debt and illiquid savings, utilizing a **gold-purchase placebo test** to isolate mechanical payroll deductions from behavioral self-control.
- Estimated **weighted, household-clustered regressions** and a **permanent/transitory income decomposition**, demonstrating consumption behavior aligns with the **Permanent Income Hypothesis**.

Gold Price Forecasting Using Time Series Model

Mar 2026 – May 2026

Term Project

Prof. Buddhananda Banerjee

- Developed a **time series forecasting model** to predict monthly gold prices using a **Kaggle dataset** covering years from 1950 to 2020.
- Applied statistical models including **GARCH, Holt-Winters, SARIMA** and **Moving Averages** to evaluate forecasting performance.
- Used **statsmodels, ARCH, and scikit-learn** libraries for stationarity checks, structural break detection, and volatility modeling.
- Achieved a lowest **MAPE of 1.23%** using a 2-point moving average model, significantly outperforming multiple statistical baselines.

Sentiment-Driven Alpha Generation for Indian Equities

Jan 2026 – Apr 2026

Bachelor's Thesis Project

Prof. Pawan Goyal

- Engineered a **real-time data pipeline** that automatically fetches financial news from nine sources and stores it in a **DuckDB database** for **50 Nifty stocks**.
- Processed text data using the **FinBERT model** to extract **sentiment scores** and combined them with technical indicators to create **17-dimensional feature vectors**.
- Developed an **LSTM and Transformer ensemble** machine learning model featuring a **custom directional loss function** to predict **overnight stock returns**.

Technical Skills

Languages & Tools: Python, C/C++, SQL (DuckDB), MATLAB, Stata, Git, Linux, Jupyter, LaTeX

Quant Finance & Math: Alpha Generation, Derivatives (Black-Scholes, Greeks), Portfolio Optimization, Risk Management, Stochastic Calculus, Monte Carlo Simulations

ML, Stats & Time Series: Deep Learning (LSTM, Transformers), FinBERT, Ensembles, scikit-learn, SARIMA, GARCH, Holt-Winters, ARCH, Structural Breaks Detection

Relevant Coursework

Market Microstructure, Financial Mathematics, Financial Econometrics, Derivatives and Risk Management, Big Data Analysis, Advanced Calculus, Linear Algebra, Numerical And Complex Analysis, Transform Calculus, Theory of Partial Differential Equations, Algorithms I*, Probability and Statistics, Regression and Time Series Model, Machine Learning, Soft Computing, Financial Institutions and Markets, Programming and Data Structures* (* courses with lab)

Awards and Responsibilities

- Honored with the **Shyamal and Sunanda Ghosh Endowment Scholarship**, granted to top-performing Engineering students.
- Two-time Qualifier for **Quantfest (2025, 2026)**, an elite algorithmic trading challenge partnered with industry leaders including **Jane Street, Optiver, and IMC**.
- Earned **Codeforces Expert** title with a peak rating of **1665**, ranking **594/24k+** (**top 2.5%**) in Round 1034 (Div. 3) (Handle: Alabantara).
- Nominated as **Departmental Representative** at IIT Kharagpur's **Career Development Center**, managing placement operations for over **900 students**.